

## Business Scenario

A marketing company is conducting an A/B test to evaluate the results of their recent marketing campaign.

An A/B test is a randomized experimentation process wherein two or more versions of a variable (web page, page element, banner, etc.) are shown to different segments of people at the same time to determine which version leaves the maximum impact and drive business metrics.

Here, the comparison is between customer groups who were presented an ad and those who were presented a PSA(Public Service Announcement). The purpose of this A/B test is to determine

- whether the Ad Campaign was successful?
- If there is a significant difference between the targeted groups?

Data dictionary:

- Index: Row index
- user id: User ID (unique)
- test group: If "ad" the person saw the advertisement, if "psa" they only saw the public service announcement
- converted: If a person bought the product then True, else is False
- total ads: Amount of ads seen by person
- most ads day: Day that the person saw the biggest amount of ads
- most ads hour: Hour of day that the person saw the biggest amount of ads

## Data Loading and Exploration

```
In [ ]: import numpy as np  
import pandas as pd
```

```
In [2]: campaign_df = pd.read_csv('/kaggle/input/datasets/faviovaz/marketing-ab-testing/marketing_AB.csv')
```

```
In [3]: # first 5 rows of the dataset
campaign_df.head()
```

```
Out[3]:
```

|   | Unnamed: 0 | user id | test group | converted | total ads | most ads day | most ads hour |
|---|------------|---------|------------|-----------|-----------|--------------|---------------|
| 0 | 0          | 1069124 | ad         | False     | 130       | Monday       | 20            |
| 1 | 1          | 1119715 | ad         | False     | 93        | Tuesday      | 22            |
| 2 | 2          | 1144181 | ad         | False     | 21        | Tuesday      | 18            |
| 3 | 3          | 1435133 | ad         | False     | 355       | Tuesday      | 10            |
| 4 | 4          | 1015700 | ad         | False     | 276       | Friday       | 14            |

```
In [4]: # meta summary of the dataset
campaign_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 588101 entries, 0 to 588100
Data columns (total 7 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Unnamed: 0      588101 non-null int64
1   user id         588101 non-null int64
2   test group      588101 non-null object
3   converted       588101 non-null bool
4   total ads       588101 non-null int64
5   most ads day    588101 non-null object
6   most ads hour   588101 non-null int64
dtypes: bool(1), int64(4), object(2)
memory usage: 27.5+ MB
```

```
In [5]: # Checking for null values
campaign_df.isnull().sum()
```

```
Out[5]: Unnamed: 0      0
user id      0
test group   0
converted    0
total ads    0
most ads day 0
most ads hour 0
dtype: int64
```

```
In [6]: # Checking for duplicated records
campaign_df.duplicated().sum()
```

```
Out[6]: np.int64(0)
```

```
In [7]: # Searching for wrongly entered data
campaign_df['test group'].unique()
```

```
Out[7]: array(['ad', 'psa'], dtype=object)
```

```
In [8]: campaign_df['converted'].unique()
```

```
Out[8]: array([False,  True])
```

```
In [9]: campaign_df['most ads day'].unique()
```

```
Out[9]: array(['Monday', 'Tuesday', 'Friday', 'Saturday', 'Wednesday',
              'Sunday',
              'Thursday'], dtype=object)
```

```
In [10]: # Removing the unnecessary column Unnamed:0

campaign_df.drop('Unnamed: 0', axis=1, inplace=True)
```

```
In [11]: # Changing data type of user id column
campaign_df['user id'] = campaign_df['user id'].astype('str')
```

```
In [12]: # Chceking min and max of 'most ads hour' column for any inconsistencies
min_hr = campaign_df['most ads hour'].min()
max_hr = campaign_df['most ads hour'].max()
print(f"Min hour : {min_hr} and Max hour : {max_hr}")
```

```
Min hour : 0 and Max hour : 23
```

## Detecting Outliers

```
In [13]: # Detecting outliers in 'total ads' column

Q1 = campaign_df['total ads'].quantile(0.25)
Q3 = campaign_df['total ads'].quantile(0.75)
IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

# Find outliers
outliers = np.where((campaign_df["total ads"] < lower) |
(campaign_df["total ads"] > upper))[0]

# Count outliers
num_outliers = len(outliers)
total_records = len(campaign_df)
outlier_percentage = (num_outliers / total_records) * 100

print(f"Total Number of records: {total_records}")
print(f"Number of outliers in 'total ads': {num_outliers}")
print(f"Percentage of outliers: {outlier_percentage:.2f}%")
```

```
Total Number of records: 588101
Number of outliers in 'total ads': 52057
Percentage of outliers: 8.85%
```

```
In [14]: # Double checking -- The 10 highest values in 'total ads' column
campaign_df['total ads'].nlargest(10)
```

```
Out[14]: 26742      2065
         47005      1778
         7187      1680
          506      1632
        15334      1491
        16525      1414
         9605      1398
        80496      1391
        27800      1354
          107      1328
         Name: total ads, dtype: int64
```

As per the above results, the users are receiving more than 1000 ads in a day, which is practically impossible. And there are 50000+ outliers which is 8.85% of the total entries. Many of these are impractical values which can distort our test results.

Hence, I have decided to remove the outliers.

```
In [15]: # Removing the outliers
campaign_df = campaign_df.drop(outliers, axis=0)
```

```
In [16]: campaign_df.shape
```

```
Out[16]: (536044, 6)
```

## Hypothesis Formulation

This A/B test answers the following question -

Is there a statistically significant difference in conversion rates between ad and PSA groups?

Conversion rate is the proportion of converted users to the total number of people the ad was presented to.

- Null Hypothesis : The conversion rate is same for both Ad and PSA group.
- Alternative Hypothesis : The conversion rates of Ad group are higher than PSA group.

Statistical Test Used - Two Proportion Z-Test

- The two-proportion Z-test (also called the two-sample proportion Z-test) is a statistical hypothesis test for assessing whether two groups differ in the proportion of a binary outcome, in such a significant way that is beyond chance.

Reasons for choosing this test:

- There are 2 independent group Ad and PSA.
- The outcome is binary.
- The sample size of dataset is 536,044 records which is quite large.

```
In [23]: # importing required functions for ztest and confidence interval
         from statsmodels.stats.proportion import proportions_ztest,
         confint_proportions_2indep
```

```
In [19]: # calculating the conversions and count of each group
summary = campaign_df.groupby("test group")["converted"].agg(
    conversions="sum",
    total_users="count"
)
summary
```

Out[19]:

|            | conversions | total_users |
|------------|-------------|-------------|
| test group |             |             |
| ad         | 6889        | 514716      |
| psa        | 226         | 21328       |

```
In [20]: # extracting all 4 variables from the summary
ad_converted = summary.loc["ad", "conversions"]
ad_total = summary.loc["ad", "total_users"]

psa_converted = summary.loc["psa", "conversions"]
psa_total = summary.loc["psa", "total_users"]
```

```
In [24]: # conducting the two proportions z-test
count = np.array([ad_converted, psa_converted])
nobs = np.array([ad_total, psa_total])

z_stat, p_value = proportions_ztest(count, nobs)

print("Z-statistic:", z_stat)
print("P-value:", p_value)
```

```
Z-statistic: 3.4859086419622503
P-value: 0.000490468476045044
```

Significance level -> 0.05. It means we're willing to accept a 5% chance of making a Type I error (rejecting the null hypothesis when it is actually true). It means 5% risk of a false positive.

If  $p\_value < 0.05$ , reject the null hypothesis.

If  $p\_value \geq 0.05$ , fail to reject the null hypothesis.

```
In [25]: # Significance level 0.05
alpha = 0.05

if p_value < alpha:
    print("Reject H0")
    print("There is a statistically significant difference in conversion
rates.")
else:
    print("Fail to Reject H0")
    print("No statistically significant difference found.")
```

```
Reject H0
There is a statistically significant difference in conversion
rates.
```

### 95% Confidence Interval

The 95% confidence interval (CI) estimates the range within which the true difference in conversion rates between the Ad and PSA groups is likely to lie.

```
In [27]: ci_low, ci_high = confint_proportions_2indep(
    ad_converted,
    ad_total,
    psa_converted,
    psa_total,
    method="wald"
)

print("95% Confidence Interval:")
print((ci_low, ci_high))
```

```
95% Confidence Interval:
(np.float64(0.001378112745129874),
 np.float64(0.004197248584070076))
```

Since both `ci_low` and `ci_high` are above 0, we can say that difference in conversion rates between Ad and PSA group is significant.

```
In [ ]:
```